

Project Overview:

The Customer Churn Prediction and Analysis project involves building a machine learning model to predict customer churn. This project utilizes data science techniques such as data collection, exploration, feature engineering, machine learning, and model deployment, with the goal of identifying customers at risk of leaving and enabling the company to take proactive retention measures.

Milestone 1: Data Collection, Exploration, and Preprocessing

Objectives:

- Collect, explore, and preprocess customer churn data to prepare for analysis and model building.

Deliverables:

- EDA Report: A document summarizing key insights from data exploration and preprocessing decisions.
- Interactive Visualizations: An EDA notebook showcasing visualizations that reveal key patterns and relationships.
- Cleaned Dataset: A dataset that is cleaned and prepared for machine learning.

Dataset source:

<https://www.kaggle.com/datasets/blastchar/telco-customer-churn/>

Extract:

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes	No
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes
6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No	No	No
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes

Content

Each row represents a customer, each column contains customer's attributes described on the column Metadata.

The data set includes information about:

- Customers who left within the last month – the column is called Churn
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges
- Demographic info about customers – gender, age range, and if they have partners and dependents

1. Objective

Milestone 1 focused on collecting, exploring, cleaning, and preparing the Telco Customer Churn dataset for deeper analysis and future machine learning modeling.

2. Dataset Summary

Item	Result
Dataset	Telco Customer Churn
Records	7043
Original Columns	21
Duplicate Records	0
Churn Rate	26.5% churned / 73.5% retained
Cleaned Dataset Shape	7043 rows x 21 columns

3. Data Exploration Completed

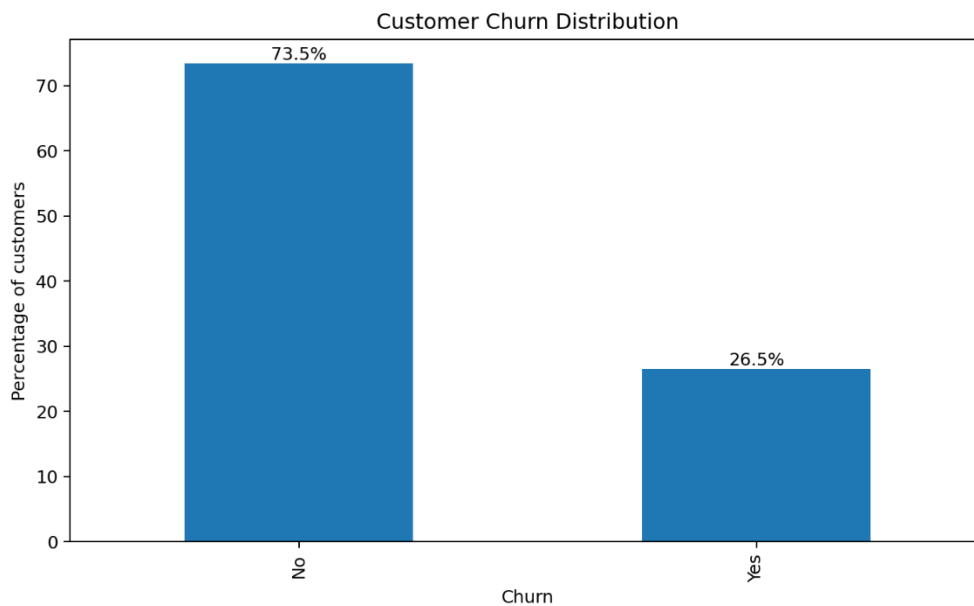
Inspected dataset shape, data types, and feature descriptions.

Checked missing values, duplicates, and basic statistical summaries.

Analyzed numerical features such as tenure, MonthlyCharges, and TotalCharges.

Analyzed categorical features such as Contract, InternetService, PaymentMethod, Partner, Dependents, and Churn.

Created visualizations including bar charts, histograms, pie charts, and boxplots.



4. Cleaning and Preprocessing Decisions

Converted TotalCharges from object/text to numeric and handled 11 missing/blank values using median imputation.

Verified that no duplicate customer records were present.

Standardized inconsistent service values such as No internet service and No phone service into No where needed for analysis/model preparation.

Reviewed outliers in tenure, MonthlyCharges, and TotalCharges without automatically removing valid customer behavior.

5. Key Insights

Overall churn rate is 26.5%, which is a substantial customer loss risk.

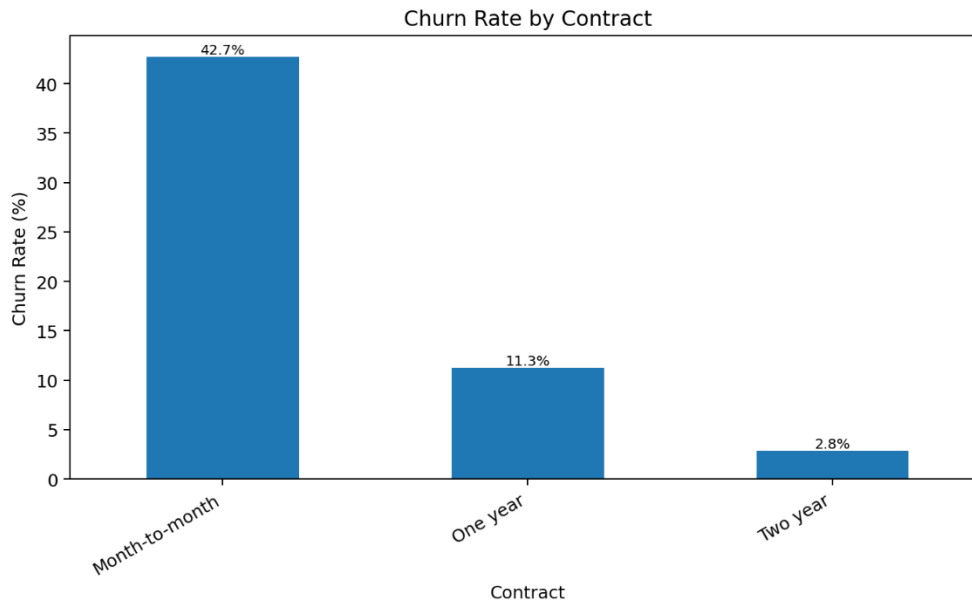
Month-to-month contract customers have a much higher churn rate (42.7%) than one-year (11.3%) and two-year customers (2.8%).

Fiber optic customers have a higher churn rate (41.9%) than DSL customers (19.0%).

Electronic check customers show the highest payment-method churn rate (45.3%).

Senior citizens show higher churn (41.7%) than non-senior customers (23.6%).

Customers with shorter tenure are more likely to churn, while longer-tenure customers are more stable.



6. Business Implications

The company should focus retention actions on new customers and month-to-month subscribers.

Pricing and service-quality review is recommended for high-charge and fiber optic segments.

Support and security services can be positioned as retention levers.

The cleaned dataset is ready for advanced analysis and feature engineering in Milestone 2.

7. Deliverables Produced

EDA report document.

EDA notebook with visualizations.

Cleaned customer churn dataset prepared for next milestones.

Milestone 2: Advanced Data Analysis and Feature Engineering

Objectives:

- Perform deeper data analysis and enhance feature selection and engineering to improve the model's predictive power.

Deliverables:

- Data Analysis Report: A comprehensive report on statistical analysis and insights derived from advanced feature analysis.
- Enhanced Visualizations: Interactive, insightful visualizations or dashboards that highlight churnrelated trends and important features.
- Feature Engineering Summary: Documentation outlining new features, transformations, and their expected impact on model performance.

1. Objective

Milestone 2 extended the initial exploration by applying statistical testing, engineered feature creation, encoding, normalization, and feature selection to prepare a machine-learning-ready dataset.

2. Advanced Statistical Analysis

Test	Features	Outcome
Independent T-Test	tenure, MonthlyCharges, TotalCharges	All tested numerical features showed statistically significant differences between churned and retained customers.
Chi-Squared Test	Contract, InternetService, PaymentMethod, OnlineSecurity, TechSupport	All tested categorical features showed strong relationships with churn.

3. Feature Engineering Summary

Engineered Feature	Purpose	Expected Impact
Tenure Groups	Segments customers by loyalty stage	Improves interpretation of early-stage vs long-term churn.
Total Services	Counts subscribed services	Measures engagement and service adoption.
Average Monthly Spending	Calculates spending relative to tenure	Highlights customer value and billing behavior.
Charge Per Service	Measures cost intensity per subscribed service	Identifies customers paying high value for low service usage.

Long-Term Customer Flag	Classifies customers with tenure >= 24 months	Captures loyalty in a simple binary form.
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4. Encoding and Normalization

Categorical variables were encoded using one-hot encoding.

Boolean dummy variables were converted to numeric 0/1 values to support ML algorithms.

Numerical features were normalized using MinMaxScaler so values remain non-negative between 0 and 1.

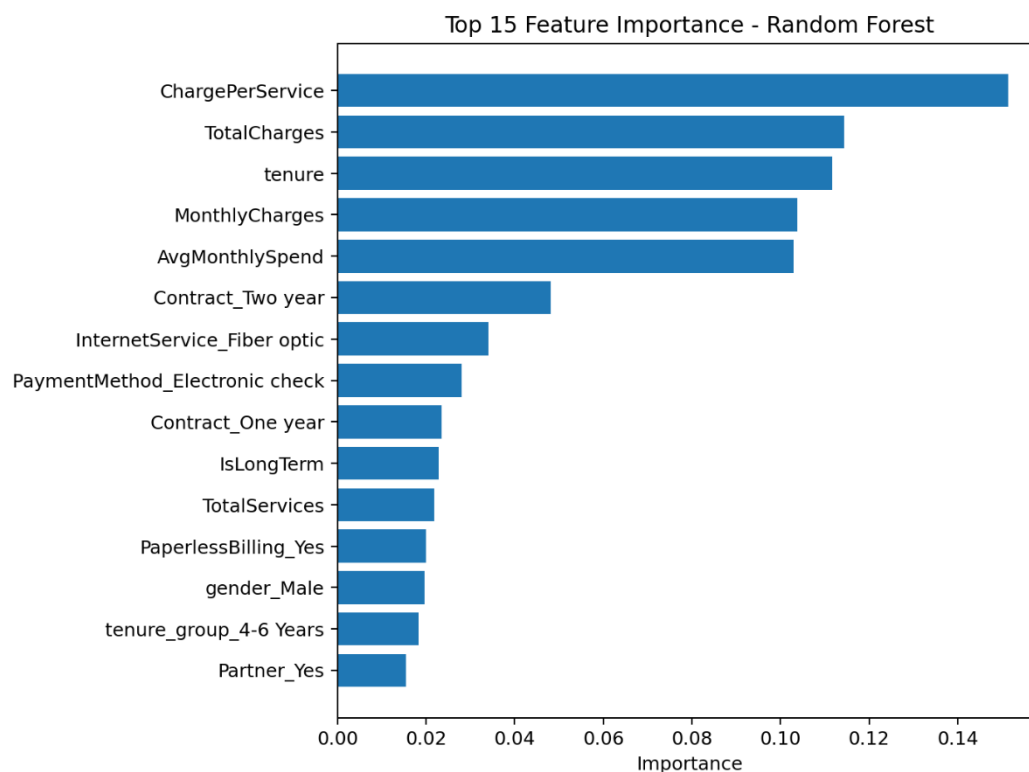
customerID was excluded from modeling because it is an identifier, not a predictive feature.

5. Feature Selection

Random Forest feature importance was used to rank predictors.

The final ML-ready selected dataset retains the top 15 most important features plus the Churn target column.

This reduces weak predictors and keeps the next modeling phase more efficient and interpretable.



6. Top 15 Selected Features

Rank	Feature
1	ChargePerService
2	TotalCharges
3	tenure
4	MonthlyCharges

5	AvgMonthlySpend
6	Contract_Two year
7	InternetService_Fiber optic
8	PaymentMethod_Electronic check
9	Contract_One year
10	IsLongTerm
11	TotalServices
12	PaperlessBilling_Yes
13	gender_Male
14	tenure_group_4-6 Years
15	Partner_Yes

7. Key Advanced Insights

Contract type remains one of the strongest churn indicators: month-to-month customers churn at 42.7%.

Payment method is highly relevant: electronic check customers churn at 45.3%.

Higher charges, lower tenure, and weaker service engagement are important churn risk signals.

Customers with more services and longer tenure show stronger retention behavior.

8. Deliverables Produced

Advanced data analysis report.

Feature engineering summary.

Full encoded and normalized dataset.

Top 15 selected-features ML-ready dataset.

Feature importance ranking CSV.

Milestone 3: Machine Learning Model Development and Optimization

Objectives:

- Build, train, and optimize machine learning models to predict churn.

Deliverables:

- Model Evaluation Report: A detailed report comparing model performance with evaluation metrics.
- Model Code: Python code used to train, optimize, and evaluate the models.
- Final Model: The best-performing churn prediction model, tuned and ready for deployment.

Sections:

1. Train / test split + SMOTE
2. Baseline training of 5 classifiers
3. 5-fold stratified cross-validation
4. Confusion matrices + classification reports
5. GridSearchCV tuning (Gradient Boosting & Random Forest)
6. Final comparison + ROC curves

1. Train / test split + SMOTE

Stratified 80/20 split preserves the 73/27 churn ratio in both subsets. SMOTE is applied ****only to the training set**** so the test set still reflects real-world distribution.

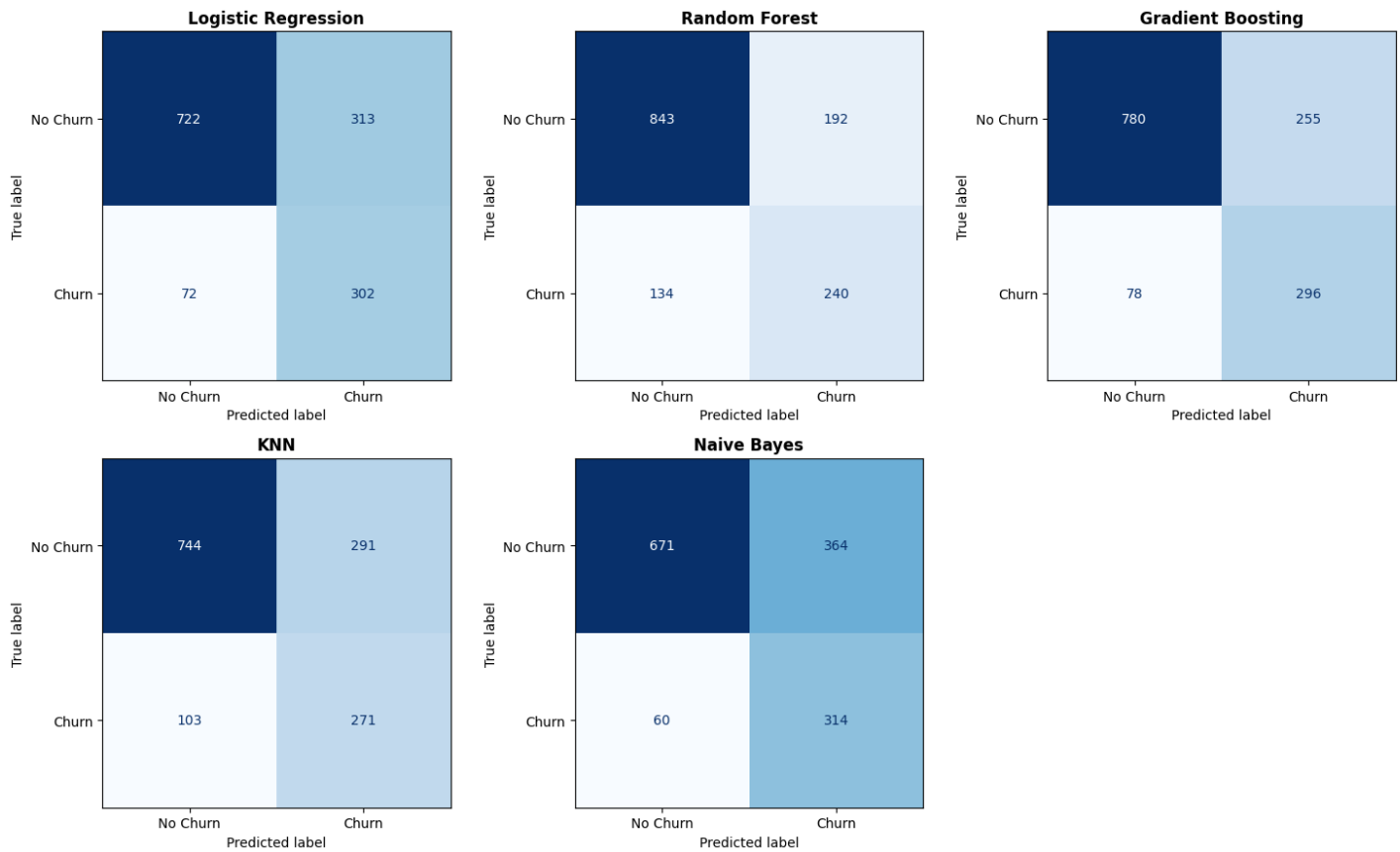
2. Baseline models — train and quick metrics

Five classifiers covering linear, ensemble, instance-based, and probabilistic approaches.

3. 5-fold Stratified Cross-Validation (SMOTE applied per fold)

Wraps SMOTE inside an `imblearn.Pipeline` so synthetic minority samples never leak into the validation fold.

4. Confusion matrices + classification reports



5. Hyperparameter tuning — GridSearchCV

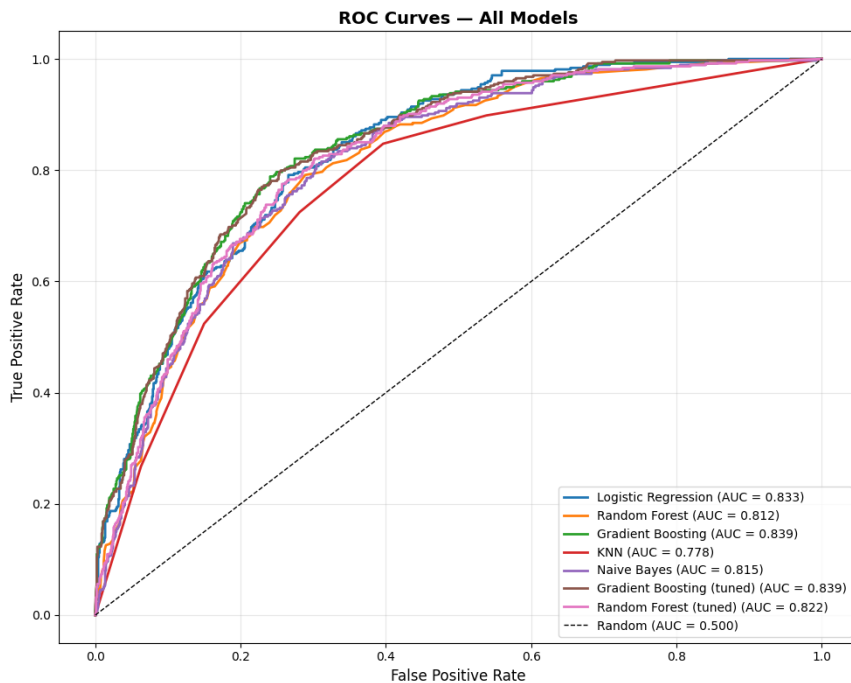
Tune the two strongest baselines: Gradient Boosting and Random Forest.

6. Final comparison — baseline + tuned, on the held-out test set

FINAL MODEL COMPARISON (sorted by ROC-AUC)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Gradient Boosting (tuned)	0.7466	0.5145	0.8075	0.6285	0.8393
Gradient Boosting	0.7637	0.5372	0.7914	0.6400	0.8392
Logistic Regression	0.7268	0.4911	0.8075	0.6107	0.8331
Random Forest (tuned)	0.7700	0.5556	0.6684	0.6068	0.8218
Naive Bayes	0.6991	0.4631	0.8396	0.5970	0.8146
Random Forest	0.7686	0.5556	0.6417	0.5955	0.8120
KNN	0.7204	0.4822	0.7246	0.5791	0.7776

ROC curves — all models on one chart



7. Pick the winner — by F1

ROC-AUC ranks the untuned GB at the top, but tuned GB has higher F1 and recall on the churn class — which matter more for a retention use case.

Best model (by F1): Gradient Boosting

Test ROC-AUC = 0.8392 | F1 = 0.64

8. Save the winning model + its metadata

We save TWO files that work as a pair:

1. The model itself -> .joblib (the actual "brain", saved with joblib)

2. Info about it -> .json (name, scores, settings, saved with json)

Model saved to: best_churn_model.joblib

Metadata saved to: best_churn_model_info.json

Milestone 4: MLOps, Deployment, and Monitoring

Objectives:

- Implement MLOps practices and deploy the churn prediction model for real-time or batch predictions.

Deliverables:

- **Deployed Model:** A fully functional API or cloud-deployed model that can make real-time churn predictions.
 - **MLOps Report:** A report detailing the MLOps pipeline, experiment tracking, model deployment, and monitoring setup.
 - **Monitoring Setup:** Documentation on how to track model performance and trigger updates or retraining.
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Milestone 5: Final Documentation and Presentation

Objectives:

- Prepare final documentation and create a presentation for stakeholders that showcases the project's results and business impact.

Deliverables:

- **Final Project Report:** A detailed summary of the project’s process, from data collection to deployment, and the business impact of churn prediction.
 - **Final Presentation:** A polished presentation for business stakeholders, explaining the model’s value and usage.
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Final Milestones Summary:

Milestone	Key Deliverables
1. Data Collection, Exploration & Preprocessing	EDA Report, Interactive Visualizations, Cleaned Dataset
2. Advanced Data Analysis, Visualization & Feature Engineering	Data Analysis Report, Enhanced Visualizations, Feature
Engineering Summary	
3. Model Development & Optimization	Model Evaluation Report, Model Code, Final Model
4. MLOps, Deployment & Monitoring	Deployed Model, MLOps Report, Monitoring Setup
5. Final Documentation & Presentation	Final Project Report, Final Presentation

Conclusion:

The Customer Churn Prediction and Analysis project aims to build a machine learning model that identifies customers at risk of leaving, helping businesses take action to retain them. This step-by-step process focuses on data exploration, feature engineering, model development, deployment, and continuous monitoring to ensure that the churn prediction system remains effective over time.